

Characterization of sub-pollen particles in size-resolved atmospheric aerosol using chemical tracers

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Supporting Information

Table S1: Correlation between meteorological conditions and tracer concentrations

	Daily maximum temperature (°C)	Daily mean temperature (°C)	Daily mean relative humidity (%)	Cumulative rainfall (mm)	Fructose (ng m ⁻³)	Sucrose (ng m ⁻³)	Glucose (ng m ⁻³)	Mannitol (ng m ⁻³)
Fructose (ng m ⁻³)	<u>0.47</u>	<u>0.52</u>	0.31	<u>0.44</u>				
Sucrose (ng m ⁻³)	<u>0.40</u>	0.28	<u>-0.41</u>	-0.18	0.11			
Glucose (ng m ⁻³)	<u>0.55</u>	<u>0.63</u>	<u>0.47</u>	0.31	<u>0.75</u>	-0.02		
Mannitol (ng m ⁻³)	<u>0.53</u>	<u>0.62</u>	<u>0.38</u>	0.30	<u>0.72</u>	-0.04	<u>0.90</u>	
Endotoxin (EU m ⁻³)	0.31	0.34	0.34	0.42	<u>0.65</u>	0.13	0.45	0.37

Bold: $p < 0.05$, bold and underline: $p < 0.01$

Table S2: Estimation of ambient concentrations of SPP in PM_{2.5}, the number of ruptured pollen grains, and percent of pollen ruptured for 5 rainy days in 2019 using fructose-to-pollen mass ratio of oak pollen grains

Date	Background subtracted PM _{2.5} fructose concentration (ng m ⁻³)	Estimated SPP mass concentration (ng m ⁻³)	Estimated SPP contribution to PM _{2.5} mass (%)	Estimated ruptured pollen (grains m ⁻³)	Estimated ruptured pollen (%)
May 8	0.4	11	0.1	1	1
May 16	1.1	32	0.3	3	1
May 17	0.04	1.4	0.01	<1	<1
May 18	4.6	140	1.5	13	11
May 24	6.1	180	2.2	17	3

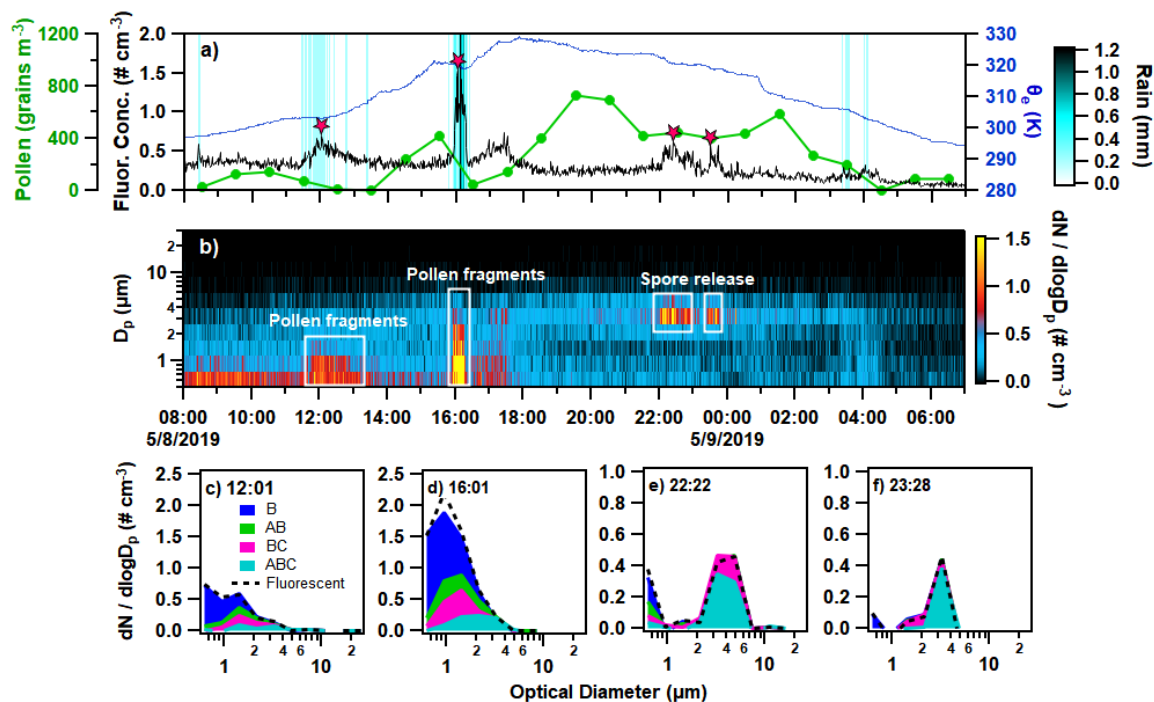


Figure S1: Co-located measurements on May 8: a) fluorescent particle number concentration (1 min averaged; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. Size-resolved number concentration of fluorescent particles at the peak of the signal at c) 12:01, d) 16:01, and spore release at e) 22:22, and f) 23:28. Measurements from 02:00 to 22:00 on May 8 have been discussed previously by Hughes et al. (2020). Figure 1, 2, and 3 shows PM_{2.5} and size resolved tracer concentrations in PM collected from May 8, 08:00 to May 9, 07:00.

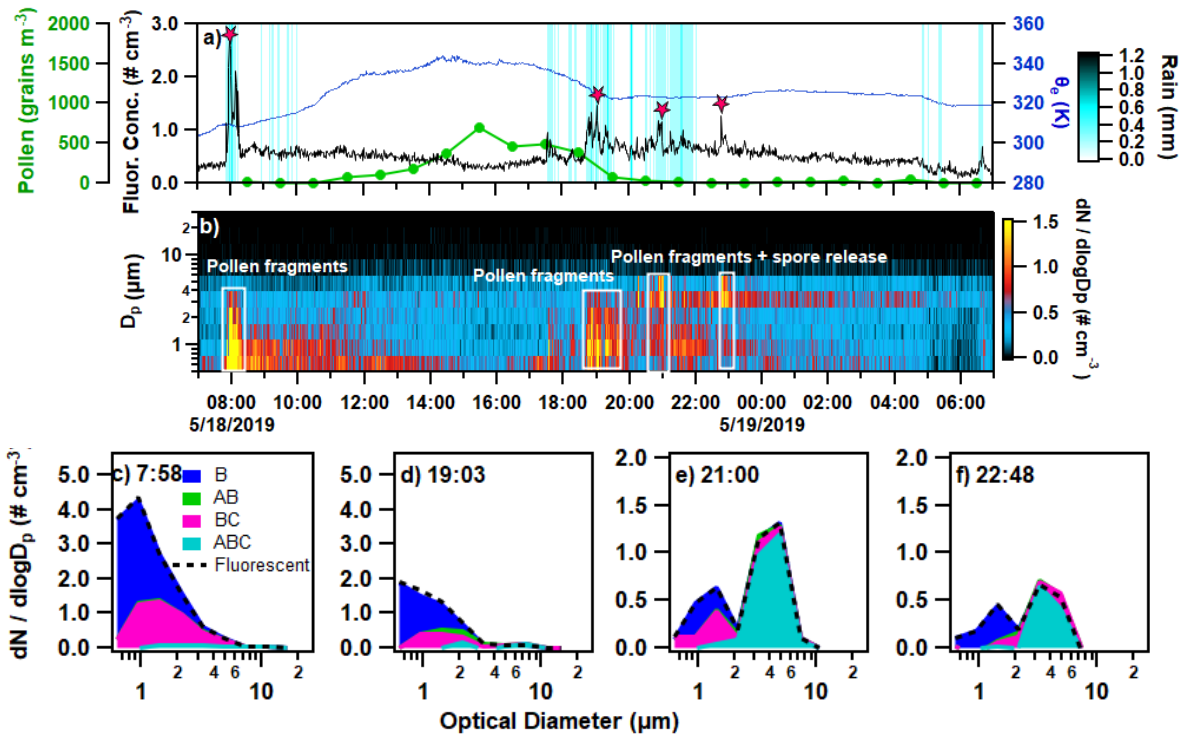


Figure S2: Co-located measurements on May 18: a) fluorescent particle number concentration (1 min averaged; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. Size-resolved number concentration of fluorescent particles at the peak of the signal at c) 7:58, d) 19:03, and spore release peak at e) 21:00, and f) 22:48. Measurements from May 18 00:00 to May 19 00:00 have been discussed previously by Hughes et al. (2020). Figure 1, 2, and 3 shows $PM_{2.5}$ and size resolved tracer concentrations in PM collected from May 18, 08:00 to May 19, 07:00.

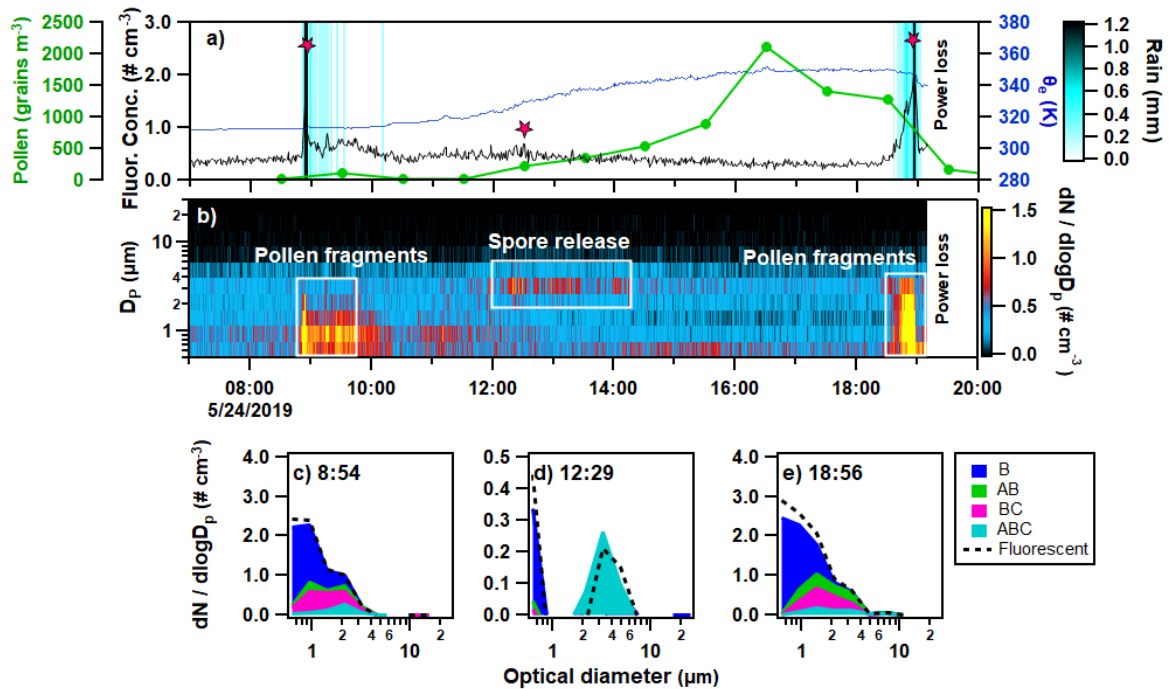


Figure S3: Co-located measurements on May 24: a) fluorescent particle number concentration (1 min; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. Size-resolved number concentration of fluorescent particles at the peak of the signal at c) 8:54, d) 12:29, and e) 18:56. Measurements from May 24 00:00 to 20:00 have been discussed previously by Hughes et al. (2020). Figure 1, 2, and 3 shows $PM_{2.5}$ and size resolved tracer concentrations in PM collected from May 24, 08:00 to May 25, 07:00.

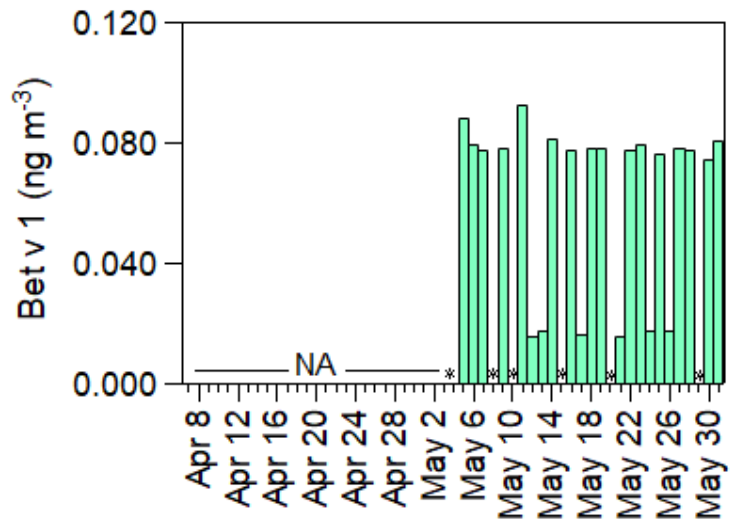


Figure S4: Bet v 1 concentrations in PM_{2.5} samples. Bet v 1 was detected in 22 of 28 PM_{2.5} samples analyzed for this allergen and resulting concentrations were close to the lowest instrument calibration standard. No consistent trends were observed in the PM_{2.5} Bet v 1 concentrations. *Not detected

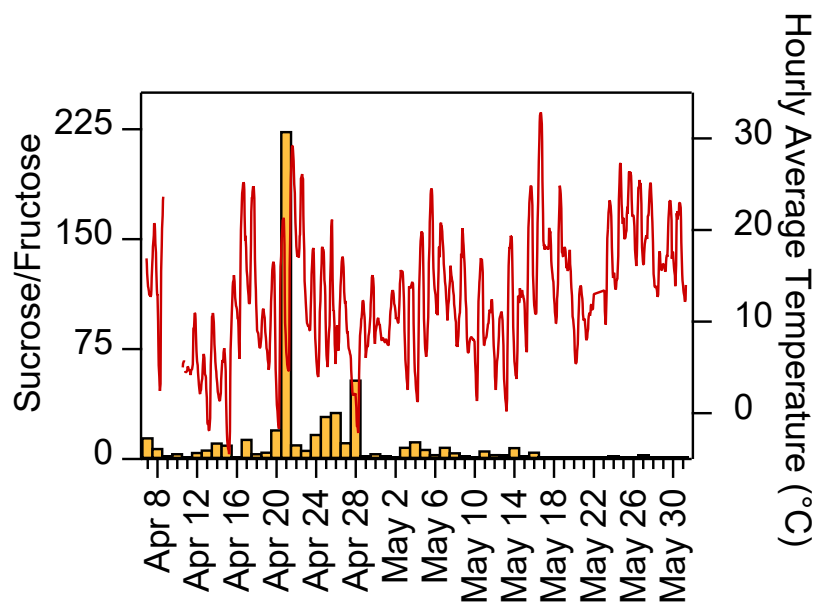


Figure S5: Sucrose/Fructose ratio and hourly average temperature.

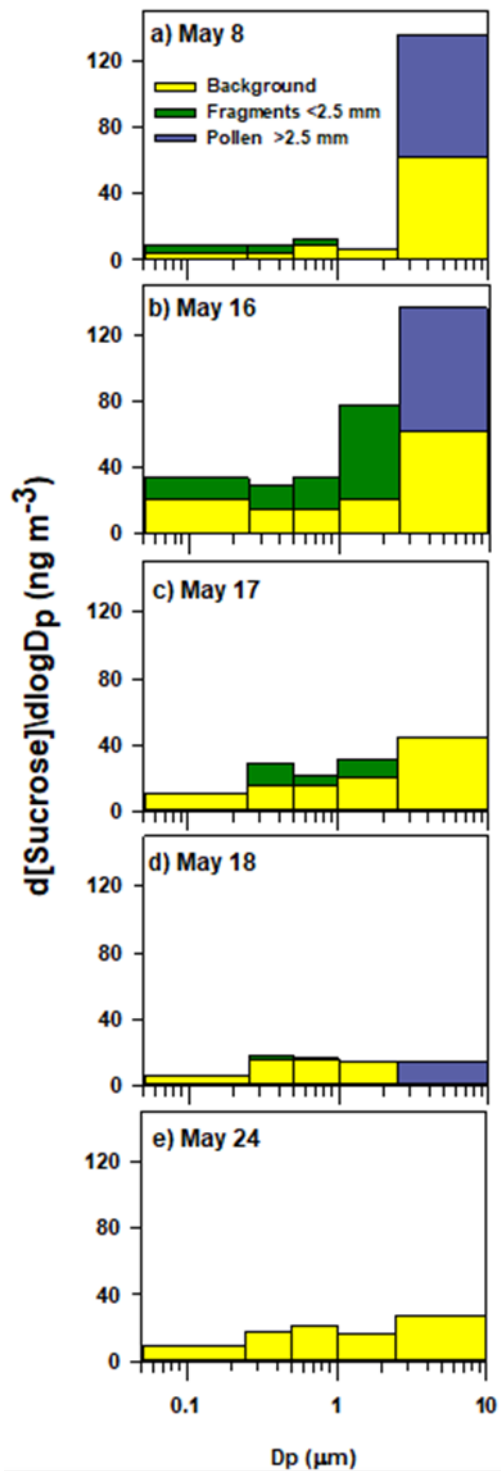


Figure S6: Mass concentrations of sucrose measured in ambient PM collected by a 5-stage impactor on 5 days with local rain events. Measurements are reported relative to background periods corresponding to dry sampling periods.

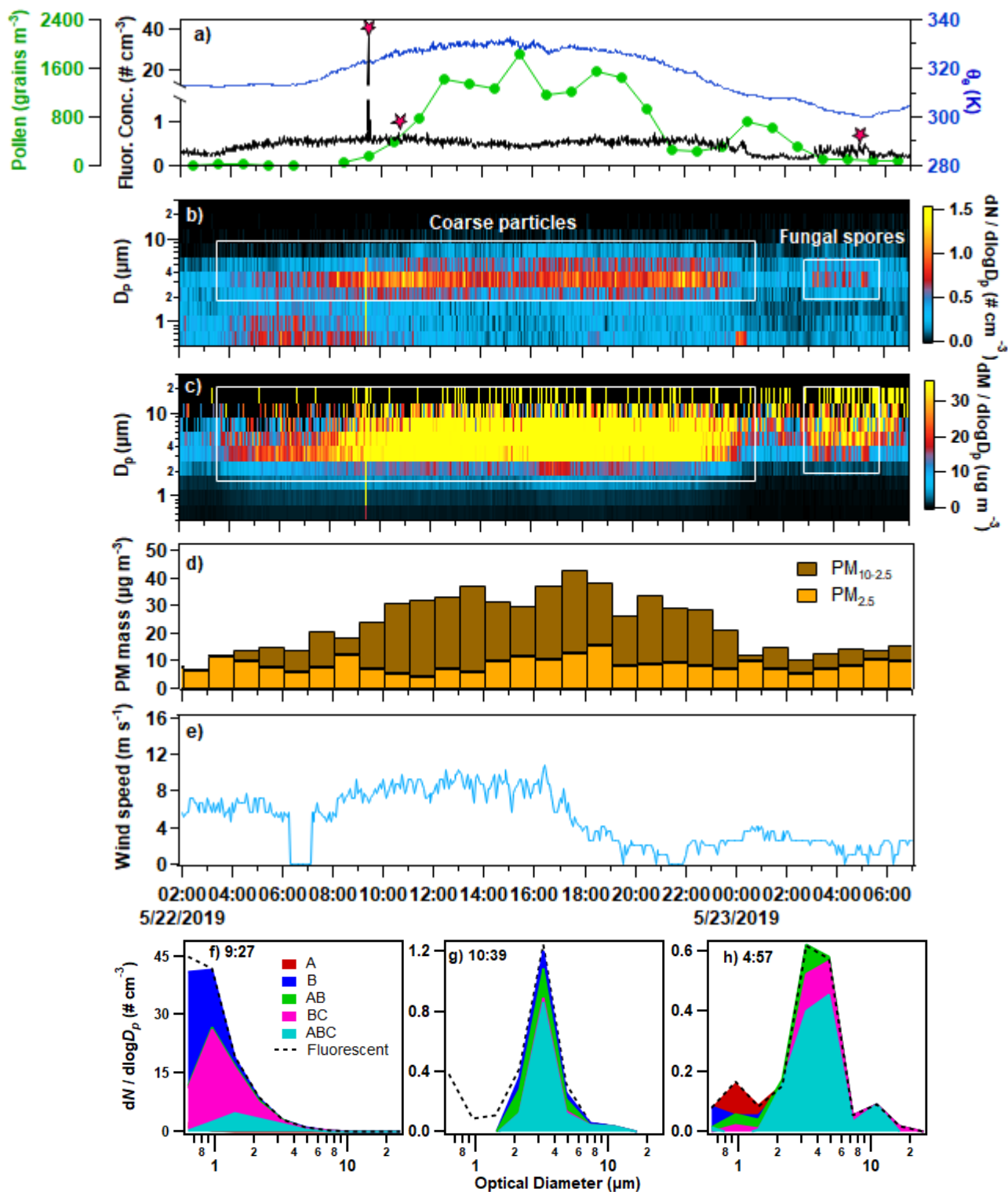


Figure S7: Co-located measurements on May 22: a) fluorescent particle number concentration (1 min; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. c) Size resolved mass concentration of fluorescent particles, d) PM mass concentration, and e) 5-min averaged wind speed data from Iowa City Municipal airport, and size-resolved number concentration of fluorescent particles at the peak of the signal at f) 9:27, g) 10:39, and h) 4:57.

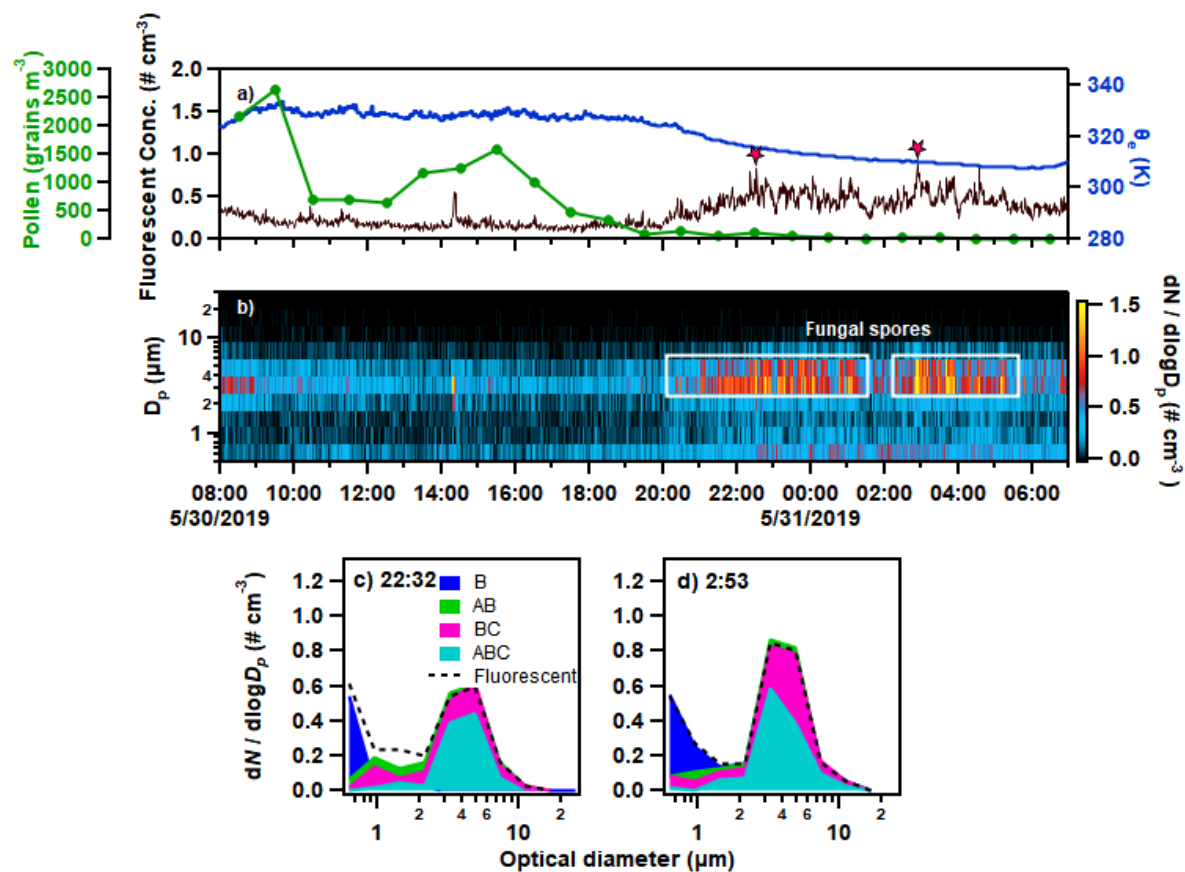


Figure S8: Co-located measurements on May 30: a) fluorescent particle number concentration (1 min; black line), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. Size-resolved number concentration of fluorescent particles at the peak of the signal for events at c) 22:32, and d) 2:53. No rain was observed on May 30. Figure 2 shows $PM_{2.5}$ tracer concentrations in PM collected from May 30, 08:00 to May 31, 07:00.

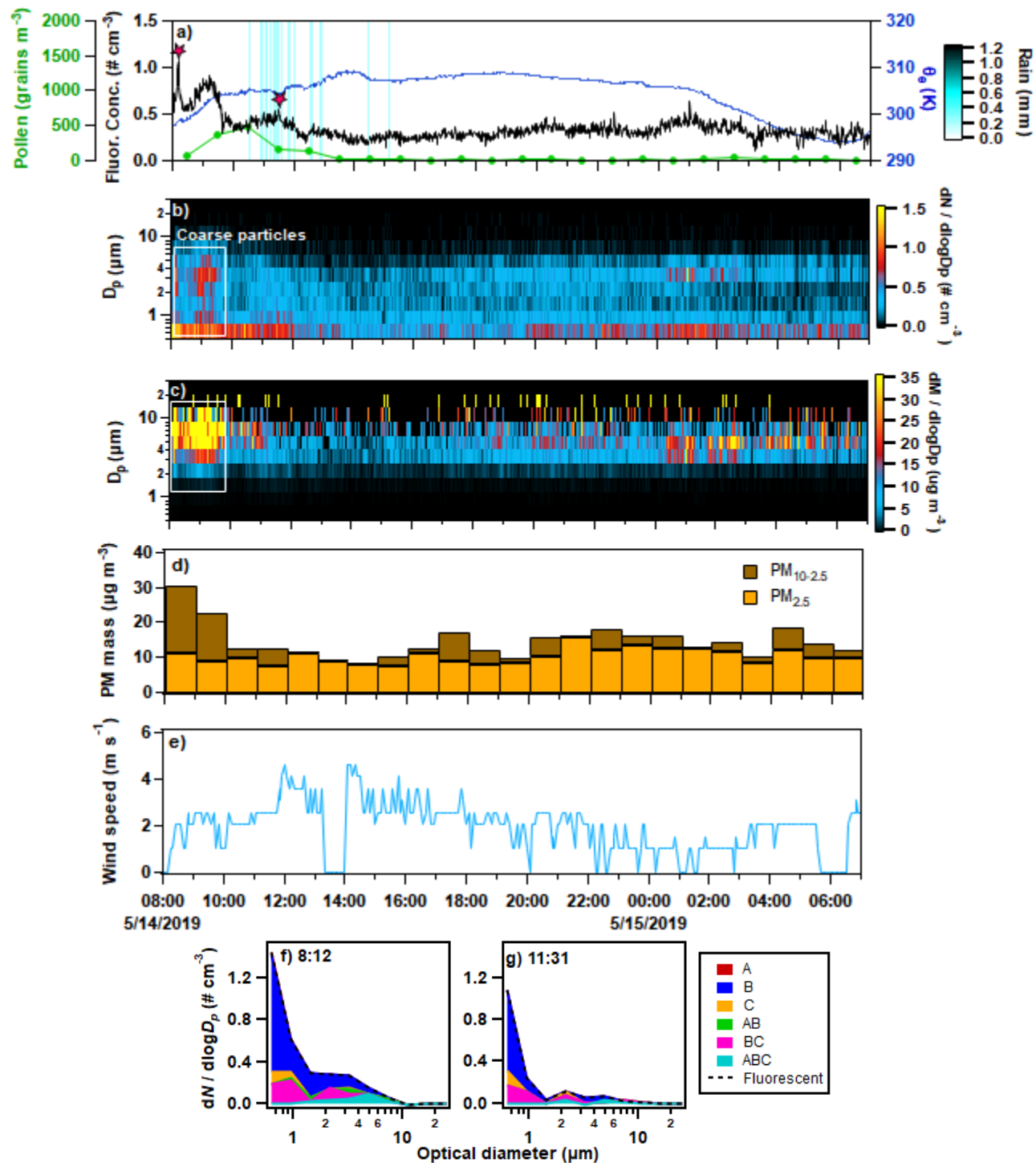


Figure S9: Co-located measurements on May 14: a) fluorescent particle number concentration (1 min; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), and b) size-resolved number concentrations of fluorescent particles. c) Size resolved mass concentration of fluorescent particles, d) PM mass concentration, and e) 5-min averaged wind speed data from Iowa City Municipal airport, and size-resolved number concentration of fluorescent particles at the peak of the signal at e) 8:12 and f) 11:31. Figure 2 shows $\text{PM}_{2.5}$ tracer concentrations in PM collected from May 14, 08:00 to May 15, 07:00.

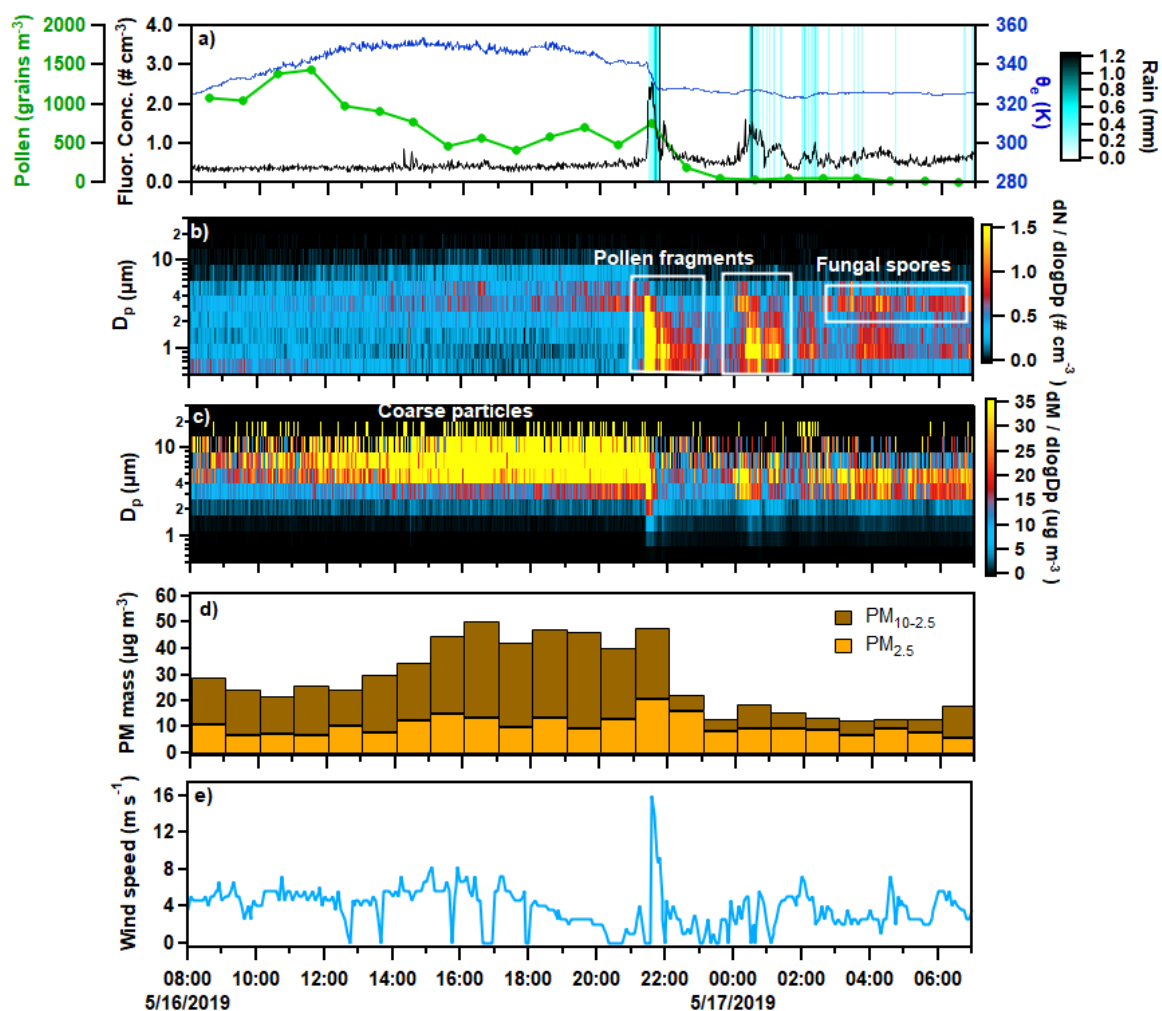


Figure S10: Co-located measurements on May 16: a) fluorescent particle number concentration (1 min; black line), rain (1 min; blue bars), intact pollen grain count (hourly; green markers and line), equivalent potential temperature (θ_e ; blue line), b) size-resolved number concentrations of fluorescent particles, c) size resolved mass concentration of fluorescent particles, d) PM mass concentration, and e) 5-min averaged wind speed data from Iowa City Municipal airport. Measurements from May 16, 16:00 to May 17, 16:00 have been discussed previously by Hughes et al. (2020). Figure 1, 2, and 3 shows $PM_{2.5}$ and size resolved tracer concentrations in PM collected from May 16, 08:00 to May 17, 07:00.

Works Cited

Hughes, Dagen D., Chamari B. A. Mampage, Lillian M. Jones, Zehui Liu, and Elizabeth A. Stone. 2020. Characterization of Atmospheric Pollen Fragments during Springtime Thunderstorms. *Environmental Science & Technology Letters* 7 (6):409-414. doi: 10.1021/acs.estlett.0c00213.